SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

CO1 - ASSESSMENT TOOL - 1

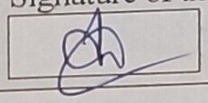
DESCRIPTIVE WRITTEN TEST

Course Code	**CSA3717**	**Course Title**	**Software Testing**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 1: Descriptive Written Test**
Weightage	**50% of CIA**	**Total Marks**	**100**
CO1 Statement	*Apply software testing principles to enhance the effectiveness of software quality assurance and improve defect detection strategies.* **(BL3)**		
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Section/Batch	**Slot A / 2023**	**Date**	**21-10-2026**

Code of Conduct:

I, Anchal Kumari, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



...eria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
...derstanding Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
...planation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
...plication	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
...ganization	20%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
...ccuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	

A banking application team is facing frequent failures due to unstructured testing practices. As a test engineer, apply the concept of testing as an engineering activity to redesign the testing approach. Explain how systematic test design, tools, and standards can be introduced. Illustrate how this approach improves defect detection and quality. Suggest suitable techniques to replace ad-hoc testing. Justify your solution with a practical example.

Problem Statement: The banking application team is facing frequent failures due to unstructured testing practices.

Testing as an Engineering Activity

Requirement Analysis → Test planning → Test design → Test execution

Understand business & risk areas

Define scope, Strategy, resources.

create structured test case & data

Use tools & automation

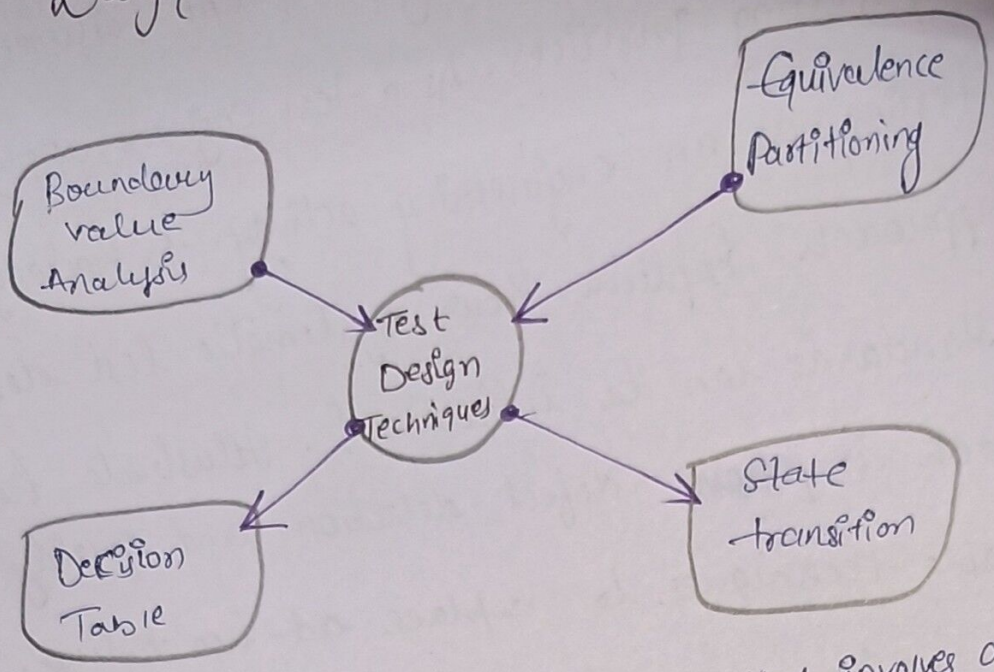
Defect tracking & reporting

Monitor, analyze, close

Continuous improvement

Metrics & feedback.

Test Design Techniques.



Software testing is an engineering activity that involves a systematic, planned, measurable, and repeatable process to identify defects and ensure that software meets its requirements.

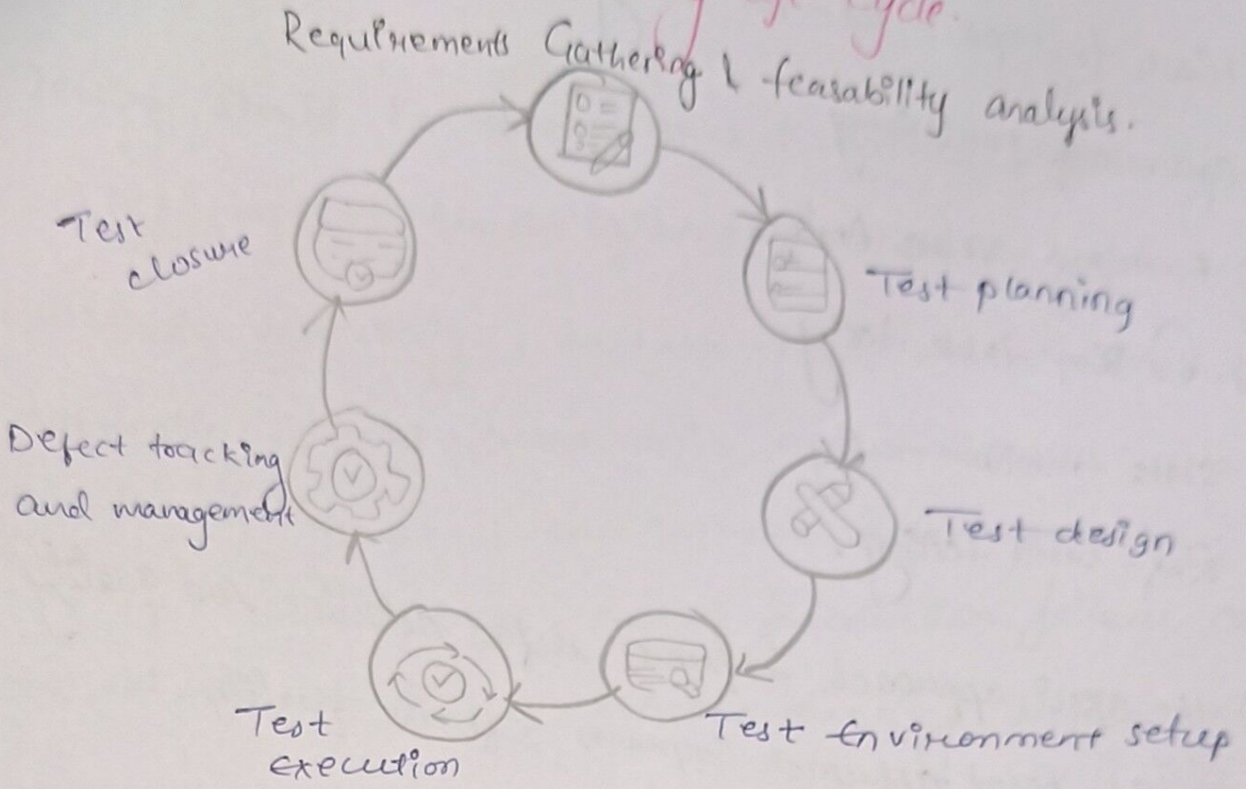
In a banking application, testing is especially important because the system handles sensitive customer information, money transfers, account balances, and financial transactions. Unstructured testing can lead to security vulnerabilities, incorrect balances, transaction failures, and loss of customer trust.

Therefore, the testing team should replace ad-hoc testing with a structured testing process that uses proven techniques, appropriate tools, and recognized standards.

Tools :-

Selenium
Postman
Jira
Jenkins
OWASP ZAP

Phases of software Testing life Cycle.



The banking team should implement the following steps:

1. Requirement analysis: Understand banking requirements, including logins, account management, fund transfers, and transaction history.
2. Test planning: Define testing scope, objectives, resources, schedule, responsibilities, and risks.
3. Test design: Create test cases, test data, expected results, and coverage criteria.
4. Test Execution: Execute manual and automated tests in a controlled environment.
5. Defect tracking: Record defects, assign priorities, fix issues and perform testing.

Techniques to replace ad-hoc testing.

(i) Equivalence partitioning.

(ii) Boundary value Analysis

(iii) Decision table testing

(iv) State transition testing

(v) Risk-based testing.

How the new approach improves defect detection and quality

The engineering based approach improves software quality in several ways:

Higher test Coverage: Requirements are mapped to test cases, reducing missed functionality.

Early defect detection: Unit and integration tests detects problems before the complete application is released.

Fewer repeated failures: Regression tests check whether new changes break previously working features.

Improved Security: Security testing identifies vulnerabilities involving authentication, authorization, and sensitive data.

Measurable results: Metrics such as test coverage, defect density, defect leakage, and pass rate helps asses quality.

Better reliability: Repeatable procedures and controlled test environments produce more consistent results.

Conclusion:

Testing should be treated as a disciplined engineering activity rather than an informal process. By implementing systematic test design, suitable tools, recognized standards, automation, risk-based testing, and continuous improvement, the banking team can improve defect detection, increase test coverage, reduce production failures, and deliver more secure and reliable banking software.